

#### Feature:

1.ESP32-S3 WROOM is designed based on ESP32S3-WROOM-1 module of PCB onboard antenna.

2.ESP32-S3 development board can be paired with OV2640/OV5640 two cameras.

3. The ESP32-S3 is a low-power MCU system-on-chip (SoC) that integrates 2.4GHz Wi-Fi and Bluetooth Low energy dual-mode wireless communication.

4.ESP32-S3 has a complete Wi-Fi subsystem and Bluetooth Low power subsystem, with industry-leading low power performance and RF performance, support a variety of low power working states, can meet the power needs of various application scenarios.

5.ESP32-S3 chip provides a wealth of peripheral interfaces, and has a variety of unique hardware security mechanisms, perfect security mechanisms enable the chip to meet the strict security requirements.

#### Performance parameters:

Controller: XtensaR dual-core 32-bit LX7 CPU up to 240MHz

#### Storage:

384 KB ROM

512 KB of SRAM

16 KB of RTC SRAM

8 MB PSRAM

Operating voltage: 3V to 3.6V.

Up to 45 GPIOs

2\* 12-bit ADC (up to 20 channels)

#### Communication interface:

Two I2C ports

2 I2S port

4 SPI interfaces

Three UART ports

1 USB oTG port.

Temperature range: -40 to 65 degrees Celsius

WiFi

Supports IEEE 802.11b /g/n protocol

Supports 20 MHz and 40 MHz bandwidth in the 2.4 GHz band

Supports 1T1R mode with data rates up to 150 Mbps

Wireless Multimedia (WMM)

Frame aggregation (TX/RX A-MPDU,TX/RX A-MSDU)

ImmediateBlock ACK ImmediateBlock ack

Fragmentation and reorganization (the Fragmentation/defragmentation) -  
Beacon automatic monitoring hardware (TSF) - 4 x virtual wi-fi interface

Support for infrastructure network (Infrastructure BSS) Station mode, SoftAP mode, and Station+ SoftAP hybrid mode Please note that when the ESP32-S3 scans in Station mode, SoftAP channels change at the same time

Antenna diversity

802.11mcFTM

Support for external power amplifiers

Bluetooth

Bluetooth LE :Bluetooth 5, Bluetoothmesh

High Power mode (20 dBm, PA shared with Wi-Fi)

The speed supports 125Kbps, 500Kbps, 1 Mbps, and 2 Mbps.

Broadcast extensions

MultipleAdvertisement Sets (Channel Selection Algorithm #2)Wi-Fi and Bluetooth coexist, sharing the same antenna

## Advanced peripheral interfaces and sensors

45 xGPIO port

### Digital interface:

- 4xSPI
- 1xLCD port (8-bit
- 16 parallel RGB, I8080, MOTO6800), support the RGB565, YUV422, YUV420, mutual transformation between YUV411
- 1xDVP 8-16 bit camera port
- 3xUART
- 2xI2C
- 2x I2S
- 1xRMT (TX/RX)
- 1x Pulse counter
- LED PWM controller with up to 8 channels
- 1x Full speed USB OTG
- 1 x USB Serial/JTAG controller
- 2x MCPWM
- 1 x SDIO host port with two card slots
- Universal DMA Controller (GDMA) with 5 receive channels and 5 transmit channels
- 1xTWAI controller, compatible with IS011898-1 (CAN Specification 2.0)

### Analog interface:

- 2x12-bit SAR ADC with up to 20 channels
- 1x Temperature sensor

-14x Capacitive sensing GPIO

Timer:

-4x54 bit universal timer

- 1 x 52-bit system timer

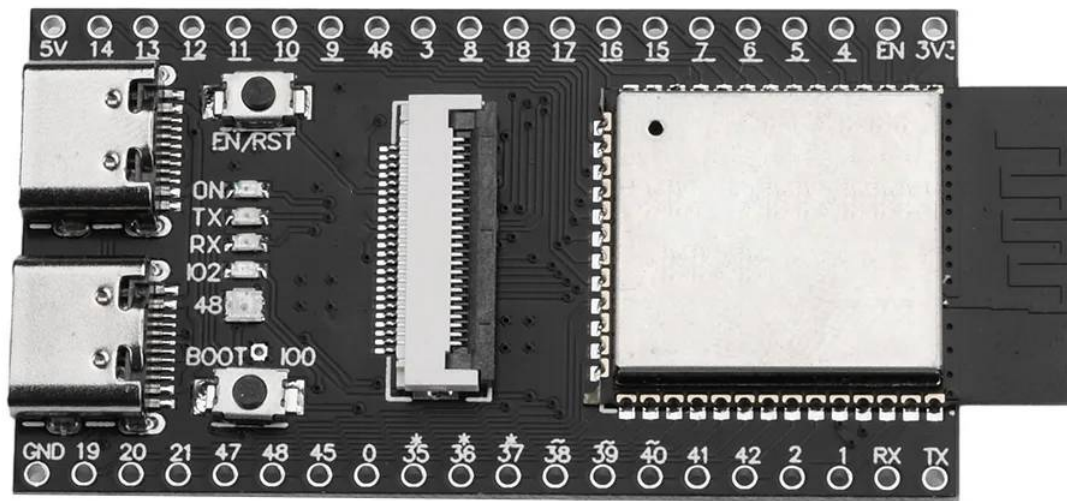
- 3x watchdog timer

Product size: 57mm\*28mm

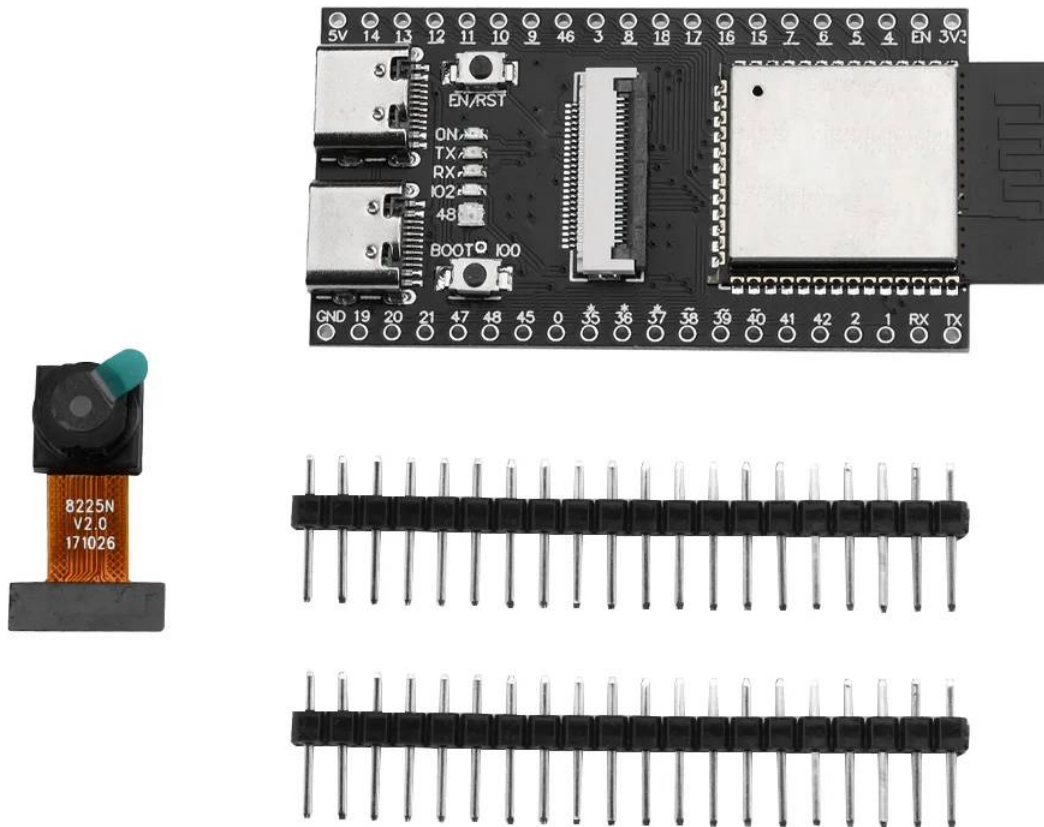
Package included:

1 x ESP32-S3 CAM Development Board 2.4G Wifi BT Module OV2640/OV5640  
Camera Module onboard ESP32-S3 WROOM-1 N16R8 Module Dual Type-C

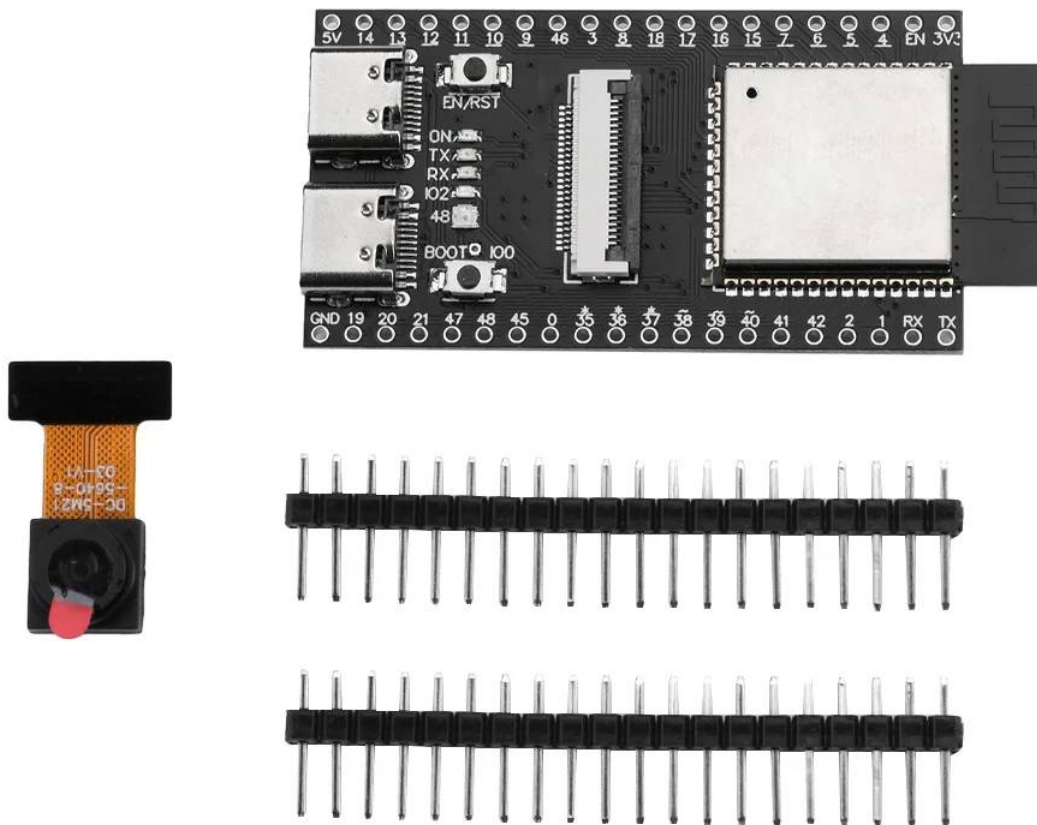
# Single module



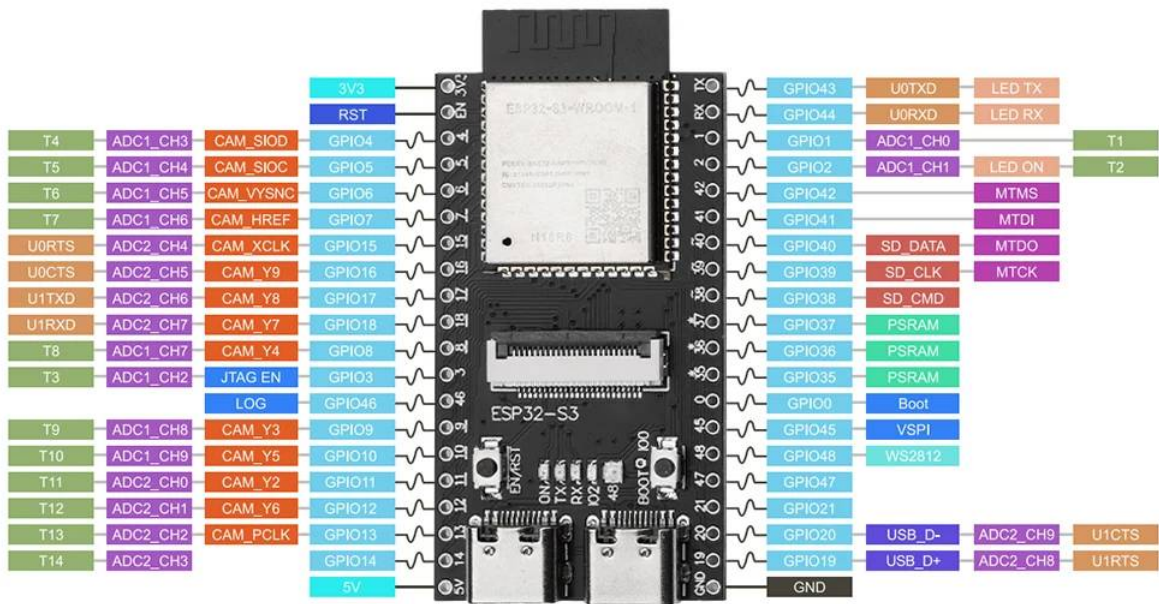
# Module + OV2640



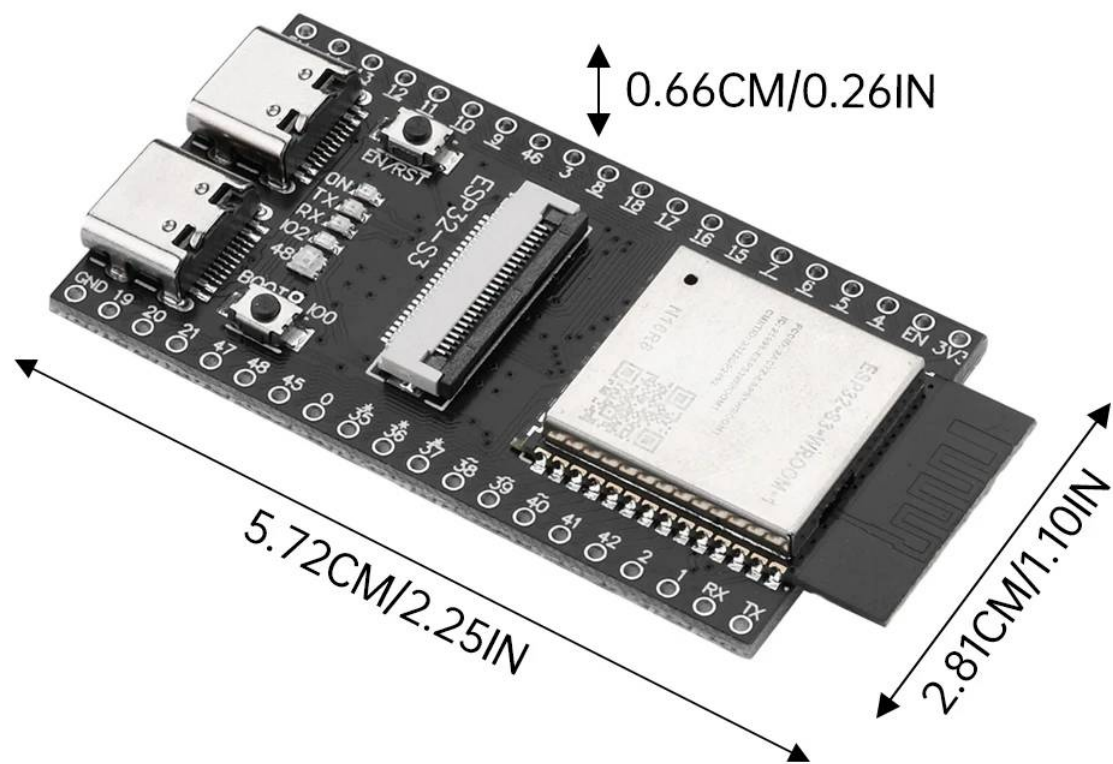
# Module + OV5640



# ESP32-S3 Pinout



|                                |                                    |                                                |                                                 |
|--------------------------------|------------------------------------|------------------------------------------------|-------------------------------------------------|
| <b>PWD</b> Power               | <b>GND</b> Ground                  | <b>Serial</b> Serial for Debugging/Programming | <b>ADCX_CH</b> Analog-to-Digital Converter      |
| <b>RESET</b> Reset the chip    | <b>GPIOX</b> GPIO Input and Output | <b>TOUCHX</b> Touch Sensor Input Channel       | <b>STRAP</b> Strapping Pin Functions            |
| <b>SD</b> On-board SD Card Pin | <b>LED</b> On-board LED Pin        | <b>WS2812</b> On-board WS2812 Pin              | <b>PSRAM</b> Built-in expansion memory chip pin |
| <b>USB</b> USB Function Pin    | <b>CAMERA</b> Camera Pin           | <b>JTAG</b> Jtag for Debugging                 | PWM Capable Pin                                 |



0.66CM/0.26IN

5.72CM/2.25IN

2.81CM/1.10IN